PREMIER UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(Project Proposal)

Application for the Approval of the Artificial Intelligence Lab Project

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Program : B.Sc. in CSE

5. Tentative Title : Stock Market Prediction Using Machine Learning

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Introduction

The stock market is a highly complex and dynamic system, where prices are influenced by a multitude of factors, including economic indicators, corporate performance, investor sentiment, geopolitical events, and market speculation. Predicting stock prices with high accuracy is a significant challenge due to the inherent volatility and the myriad of variables involved. However, accurate predictions can yield substantial financial rewards, making this an area of great interest for investors, financial analysts, and researchers alike.

Over the years, traditional methods for predicting stock prices have included statistical models, econometric techniques, and technical analysis based on historical price patterns. While these methods have provided some insights, they often fall short in capturing the nonlinear and multifactorial nature of the stock market. The advent of machine learning (ML) and artificial intelligence (AI) has introduced new possibilities for improving the accuracy of stock market predictions. These technologies can process vast amounts of data, identify complex patterns, and adapt to new information, making them particularly suited for this application.

In this project, we aim to leverage various machine learning algorithms to predict the stock prices of Netflix Inc. (NFLX), a leading company listed on the National Stock Exchange (NSE) of India. By analyzing historical stock data and applying advanced ML techniques, we seek to develop a model that can provide reliable stock price predictions. This endeavor will involve collecting data, performing exploratory data analysis (EDA), selecting relevant features, training multiple ML models, and comparing their performance to determine the most effective approach.

This proposal outlines the methodology for this project, drawing on existing research in the field. Notable studies, such as those by Vijh et al. (2020) and Ferreira et al. (2021), have demonstrated the potential of machine learning in stock market prediction, providing a strong foundation for our work. We will build upon these findings and incorporate insights from various research papers to develop and evaluate our predictive models comprehensively.

Background and Present State of the System

Stock market prediction has been a focal point of financial research for decades. Traditional approaches primarily relied on statistical methods and economic theories, which often proved insufficient in capturing the market's unpredictable nature. With the rise of machine learning, researchers have started exploring more sophisticated models that can handle the complexity of financial markets.

Recent studies have shown the effectiveness of machine learning techniques in predicting stock prices. Vijh et al. (2020) explored different ML methods for predicting stock closing prices and found models like Linear Regression and

Support Vector Machines (SVM) to be particularly effective. Ferreira et al. (2021) conducted a comprehensive review of AI applications in stock market trading, highlighting the growing importance of deep learning and neural networks in enhancing prediction accuracy. Khan et al. (2011) demonstrated that Artificial Neural Networks (ANN) could be highly accurate in predicting share prices, paving the way for more advanced neural network models.

These studies underscore the potential of machine learning in stock market prediction, suggesting that ML models can outperform traditional statistical methods by identifying complex patterns in historical data.

Objectives and Possible Outcomes

The primary objective of this project is to develop a machine learning model that can accurately predict the stock prices of Netflix company using historical data. Specific objectives include:

- **Data Collection and Preprocessing:** Uploading historical stock price data of NFLX and preprocessing it to ensure data quality.
- **Exploratory Data Analysis (EDA):** Conducting EDA to understand data characteristics and identifying significant correlations.
- **Feature Selection:** Selecting relevant features that influence stock prices, such as high, low, close prices, volume, and turnover.
- **AI Prediction Model Implementation:** Implementing and comparing various ML algorithms, including Linear Regression, Decision Trees, and Neural Networks.
- **Model Evaluation:** Evaluating models using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2) metrics through processing of data under Linear Regression library and other models.
- **Analysis and Comparison:** Providing a comprehensive analysis of the findings, comparing the effectiveness of different models in predicting stock prices.

Expected outcomes include a robust predictive model that can be used for future stock price predictions and a detailed comparative analysis of different ML techniques applied to stock market prediction.

Prediction with Machine Learning

A. Fundamental Analysis

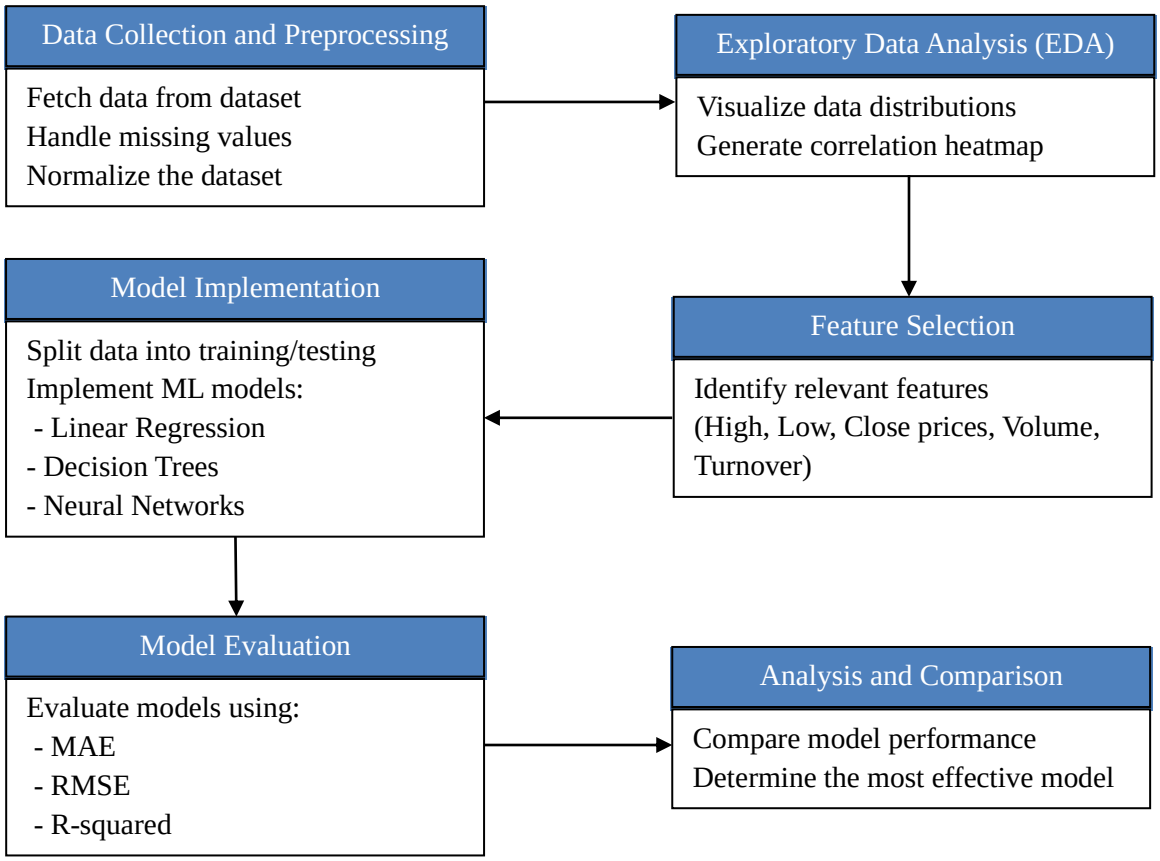
Fundamental analysis involves evaluating a stock's intrinsic value based on economic indicators, financial statements, and market conditions. This project will

consider fundamental factors such as historical stock prices, trading volumes, and turnover to inform the ML models.

B. Technical Analysis

Technical analysis focuses on historical price patterns and market trends. Indicators such as moving averages, relative strength index (RSI), and MACD will be incorporated into the ML models to enhance predictive accuracy.

Outline of Methodology



Data Collection and Preprocessing:

- Using a stock dataset [16] file of Netflix and fetch historical stock data of it.
- To preprocess the data to handle missing values and normalize the dataset.

Exploratory Data Analysis (EDA):

- To perform EDA using Pandas and Seaborn to visualize data distributions and correlations.
- Generates a heatmap to identify the correlation between different features.

Feature Selection:

- Selecting relevant features such as High, Low, Close prices, Volume, and Turnover for the prediction model.

Model Implementation:

- Splitting the dataset into training and testing sets.
- Implementing Linear Regression, Decision Tree, and Neural Network models using Scikit-learn and Keras.
- Training the models on the training set and evaluating their performance on the test set.

Model Evaluation:

- Using metrics such as MAE, RMSE, and R-squared to assess model performance.
- Comparing the results to identify the most effective model for stock price prediction.

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